SIKORSKY UH-60M BLACKHAWK FOR X-PLANE

FDCP

Flight Director Control Panel

Version 0.5.0

This manual explains the functions of the FDCP (Flight Director Control Panel) used in the “UH60M Blackhawk for X-Plane”. Following referred to as “UH60M-FDCP”.

The functionality of UH60M-FDCP is pretty close to the one used in the real Blackhawk.

We have a set of rotaries which can be used to input settings and we have a set of push buttons where we can select autopilot upper-modes. And of course we have a few displays showing us the current settings and which function the buttons have:

Since the autopilot is disabled with WOW (weight on wheels) and below 50 kias, changing upper-modes does not work while standing on ground.

We have 2 FDCP units. One for the pilot and one for the co-pilot. If one is “coupled” (CPLD), means taking control, the other is not.

The rotaries below the display can be rotated and pushed. When pushing the knob, it will take the live value from the current flight. i.e. pushing the HDG rotary will result in entering our current heading as HDG into the display. These rotaries can be controlled via commands, so it is very handy to program a 2-axis joystick rotary (if available) to select and change the values.

Pushing a button under the display will arm or de/activate the corresponding upper-mode. At the same time the unit changes to mode “CPLD”. The flight director of this unit took control.

In X-Plane only one FlightDirector is being used, but the plugin is simulating 2 FDIRs to make it work as IRL.

On the left side we also find 4 rotaries. Two 1-axis can be used to set set altimeters for the pilot and co-pilot. The upper of the other two 2-axis can be used to select the HSI source (FMS, NAV1 or NAV2) and to set the OBS indicator. The other 2-axis rotary can set the bearings for NAV1 and NAV2.

NAV1 will show up as VOR1 , LOC1 or TAC1 NEW depending of what type of nav device it is connected to. The same applies to NAV2.

On the left side of the 5 rotaries we have “GO ARND” and “LVL HOLD”

GoAround will switch to a mode which makes the aircraft climb with 700 ft/min at a speed of 70 kts. Pressing the button again or doing a ForceTrimRelease will end this mode.

LevelHold will try to hold the altitude during hover. Pressing the button will “arm” (LED=amber) it first. Once the aircraft vertical-speed is less than 0.1m/sec and air speed < 40kts, the system will engage (LED=green) and will hold the current height. Using the RALT rotary can adjust the height.

On the right side of the 5 rotaries we have “HVR POS”

HoverPos will try to keep the current position. Pressing the button will arm it first (LED=amber) and if horizontal velocity it less than 1.5m/sec and vertical speed < 900 fpm, the system will engage (LED=green) and hold the current position.

The rotaries and buttons can be mapped to NEW joystick buttons using the following commands:

uh60m/mfd/fdcp1_but_nxt	Select next button
uh60m/mfd/fdcp1_but_prv	Select previous button
uh60m/mfd/fdcp1_but_psh	Push selected button
uh60m/mfd/fdcp1_rot_nxt	Select next rotary
uh60m/mfd/fdcp1_rot_prv	Select previous rotary
uh60m/mfd/fdcp1_rot_psh	Push selected rotary
uh60m/mfd/fdcp1_rot_inc	Increment selected rotary
uh60m/mfd/fdcp1_rot_dec	Decrement selected rotary